S VEERENDRA SWAMY

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SUMMARY

Aspiring AI/ML Engineer with a strong foundation in Machine Learning, Deep Learning, Data Structures, Algorithms. Skilled in Python, C++, and Java with hands-on experience in neural networks, computer vision, and data-driven problem solving. Passionate about building intelligent systems that deliver real-world impact.

EDUCATION

Ballari Institute of Technology and Management college — **Ballari, India**
Bachelor of Technology - Artificial Intelligence and Machine Learning ; CGPA 9.23 — *2022 - 2026*

Sree Guru Thipperudra College — **India**
12th; Aggregate 80.3 — *2020 - 2021*

Narayana E-Techno School — **India**
10th; Aggregate 62 — *2019 - 2020*

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, Kotlin
Tools and Platforms: Turbo C, Visual Studio, Eclipse, Android Studio, Git
Libraries and Databases: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, MySQL, SQLite
Technologies and Frameworks: Artificial Intelligence, Machine Learning, Deep Learning, REST APIs, Flask, Streamlit, Android SDK
Additional Skills: OOP, Data Structures, Algorithms, GitHub, Agile/Scrum, Debugging

PROJECTS

EDUVID: AI-Powered Textbook to Video Platform
Tech Stack: Python, Flask, MoviePy, Hugging Face Transformers, gTTS, PDFPlumber

- Developed a platform that converts textbook PDFs into concise, engaging AI educational .
- Implemented Hugging Face Transformers for content summarization and generated high-quality voiceovers with gTTS.
- Produced videos with synchronized subtitles and relevant visual content using MoviePy.
- Designed a user-friendly Flask web interface for uploading PDFs, previewing videos, and downloading lessons.

Black and White Image Colorization Using U-Net and CNN
Tech Stack: Python, TensorFlow, Keras, OpenCV, NumPy

- Implemented a U-Net based convolutional neural network to automatically colorize grayscale images.
- Trained the model on paired grayscale and color images to learn accurate color mapping and preserve details.
- Applied image preprocessing, normalization, and augmentation to enhance model performance and generalization.
- Developed a Python interface to upload grayscale images and generate high-quality colorized outputs in real-time.

Hum2Song: Multi-track Polyphonic Music Generation from Voice Melody Transcription
Tech Stack: Python, TensorFlow, Keras, Music21, NumPy, MIDI Libraries

- Generated multi-instrument music from hummed melodies using neural networks.
- Transcribed hummed audio into note sequences with pitch detection and preprocessing.
- Predicted harmonies and accompaniments using deep learning models for polyphonic output.
- Produced MIDI/audio outputs with a Python interface for real-time conversion.

ACHIEVEMENTS

- Secured Top 3 position in HackFest coding competition by solving complex problems in Python, Data Structures, Algorithms, and AI/ML.
- Secured 2nd place in a college-level hackathon by developing innovative solutions using Python and problem-solving skills.
- Completed Google Cloud Career Launchpad Certification with a focus on Generative AI, demonstrating skills in cloud-based AI model deployment, data pipelines, and GCP services.